Eugene Tan

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Education and Appointments

University of Calgary
Postdoctoral Scholar, Department of Economics 2026–

Columbia University, School of International and Public Affairs
PhD, Sustainable Development 2026
MPhil, Sustainable Development 2025
MA, Sustainable Development 2024

Yale University, School of Forestry and Environmental Studies
MESc, Environmental and Natural Resource Economics 2019

Colorado College
BA, Environmental Science with Chemistry 2014
London School of Economics, General Course (Study Abroad) 2013

Research

Job Market Paper

Connection Charges and the Private Provision of Utility Infrastructure

Utilities expanding their networks to connect new users typically charge them either a flat fee or a distance-based connection charge. Under the latter, the first user to connect in a neighborhood funds a network extension that lowers their neighbors' connection charges. This externality creates a dynamic private provision game in which neighbors have strategic incentives both to coordinate and share infrastructure costs, and to delay adoption to free-ride on a neighbor's anticipated extension. I reconstruct the expansion of 177 electricity distribution networks in rural Rwanda. Using their evolving geography, I measure and instrument anticipated reductions in distance to the grid to estimate how strategic incentives affect adoption and cost incidence. Anticipating a 10% reduction in distance to the grid over the next eight weeks lowers the weekly probability of connecting by 0.91 percentage points, providing evidence of strategic delay. Among neighbors who connect within eight weeks of each other, the share connecting jointly rises from 32% to as much as 81% as they share more infrastructure. Customers with higher willingness to pay choose to connect earlier and pay higher charges, generating an elasticity of realized connection charges to willingness to pay of up to 0.24. Distance-based connection charges therefore create competing incentives for strategic delay and cooperation, while inducing endogenous price discrimination through the order in which customers choose to connect.

(latest version available here)

Work in Progress

The Impact of the EU's Carbon Border Adjustment Mechanism on Developing Countries

Balancing Climate and Industrial Objectives when Pricing Dirty Upstream Goods

Graphical Activity Analysis

The Energy System as a Production Network

Pre-Doctoral Publications

Changing Oils, Changing Management (with Deborah Gordon and Jeffrey Feldman) *Journal of Industrial Ecology*, 2016

Energy Return on Investment (EROI) for Forty Global Oilfields Using a Detailed Engineering-Based Model of Oil Production (with Adam Brandt, Yuchi Sun, Sharad Bharadwaj, David Livingston, and Deborah Gordon) *PLOS ONE*, 2015

Shale's True Contribution to the Oil Market (with David Livingston) *Carnegie Papers*, 2015

The Abundance of Oils in the Water-Stressed Rockies (with Deborah Gordon and Katherine Garner) *Carnegie Papers*, 2015

Conference Presentations

2026: Northeast Workshop on Energy Policy and Environmental Economics (Boston College)

$\leq$ **2025:** Climate Pipeline Project (Harvard University); Northeast Workshop on Energy Policy and Environmental Economics (Cornell University); Eastern Economic Association Annual Conference (Association of Environmental and Resource Economists); Heartland Workshop (University of Illinois Urbana-Champaign); Summer School in Environmental and Energy Economics (UC Berkeley/Sloan)

Teaching

Acting Instructor, Columbia SIPA

Economics of Energy	2023
Extraordinary Teaching Award, SIPA Energy and Environment Concentration	2023

Teaching Assistant, Columbia SIPA

Climate Change Policy and Diplomacy	2023, 2025, 2026
Advanced Economic Development	2024
Earth Systems and Environmental Politics	2021
Intermediate Microeconomics	2021
Empirical Analysis of Energy Markets	2020

Teaching Fellow, Yale University Department of Mathematics

Ordinary and Partial Differential Equations	2017, 2018

Teaching Assistant, Colorado College Environmental Program

Air: Atmospheric Chemistry and Physics	2014
Energy: Energetics and Thermodynamics	2013

Fellowships and Awards

Research Grant, Center for Political Economy, Columbia University	2025
Exploratory Research Grant, Private Enterprise Development in Low-Income Countries	2023
Travel Grant, International Growth Centre	2023

Prior Work Experience

Research Assistant, Prof. Jeffrey Shrader, Columbia University	2022
Consultant, World Bank Global Facility for Disaster Reduction and Recovery	2019

Research Assistant, Prof. William Nordhaus, Yale University	2018
Data Scientist, Center for Applied Data Science	2016–2018
Junior Fellow, Carnegie Endowment for International Peace	2014–2015

Service

Lead Organizer, SusDever, Columbia SIPA Sustainable Development	2024
Organization Committee, Interdisciplinary PhD Workshop in Sustainable Development	2023
Retreat Organizer, Sustainable Development PhD Program, Columbia SIPA	2023
PhD Student Committee, Columbia University	2021
Producer, Carnegie International Nuclear Policy Conference	2015

Citizenship: Malaysia
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